

Linux Commands Assignments - II

1. List the directory contents date wise sorted. (man ls).

Answer :

- Newest first (descending by time):

Bash : ls -lt

- Oldest first (ascending by time):

Bash: ls -ltr

```
root@PRAKHAR-96378-D1:/mnt/d/Documents/Learning# ls
12th      Finance
'3rd sem' 'Front end Development' SATI
'4th sem' 'Front end Development me' Smart-India-Hackathon
'5th sem' GitHub 'Web-D A Yu'
'6th sem' HackIndore 'Web-D CWH'
7th-sem 'Hacker House Goa Dapp' 'Web-D_Cohort'
8th-Sem 'IBM FrontEnd Skillsbuild Internship' c
'Assignments 4th sem' Interviews c++_love_babber
'Back end Development ApnaClg' JavaScript cpp
Backend-Development-me Kotlin demofile.txt
C-CAT Level-SuperMind-Hackathon java
CDAC 'ML Roadmap - Notes.pdf' 'java code'
'Clone community-classroom-repo' Projects learngit
'Custom Office Templates' 'Projects(Other'\''s)'' python
'Cyber Security' Reasoning 'web-D love babber'
DataBase 'Rohit Devaliya Python Lab Assignment.zip'

root@PRAKHAR-96378-D1:/mnt/d/Documents/Learning# ls -lt
total 92
drwxrwxrwx 1 root root 4096 May 19 15:52 CDAC
drwxrwxrwx 1 root root 4096 Apr 12 16:10 SATI
drwxrwxrwx 1 root root 4096 Mar 26 22:37 c++_Love_babber
drwxrwxrwx 1 root root 4096 Mar 18 13:45 7th-sem
drwxrwxrwx 1 root root 4096 Jan 21 17:41 Backend-Development-me
drwxrwxrwx 1 root root 4096 Jan 5 13:55 C-CAT
drwxrwxrwx 1 root root 4096 Oct 6 2025 Projects
drwxrwxrwx 1 root root 4096 Sep 5 2025 Interviews
drwxrwxrwx 1 root root 4096 Aug 10 2025 'Projects(Other'\''s)''
drwxrwxrwx 1 root root 4096 Apr 26 2025 8th-Sem
drwxrwxrwx 1 root root 4096 Jan 31 2025 GitHub
drwxrwxrwx 1 root root 4096 Jan 8 2025 Level-SuperMind-Hackathon
drwxrwxrwx 1 root root 4096 Dec 10 2024 '5th sem'
drwxrwxrwx 1 root root 4096 Dec 2 2024 '6th sem'
drwxrwxrwx 1 root root 4096 Sep 11 2024 'Front end Development me'
drwxrwxrwx 1 root root 4096 Sep 4 2024 Smart-India-Hackathon
drwxrwxrwx 1 root root 4096 Aug 14 2024 'Back end Development ApnaClg'
drwxrwxrwx 1 root root 4096 Aug 11 2024 c
drwxrwxrwx 1 root root 4096 Aug 3 2024 Reasoning
drwxrwxrwx 1 root root 4096 Aug 2 2024 'IBM FrontEnd Skillsbuild Internship'
drwxrwxrwx 1 root root 4096 Jul 28 2024 HackIndore
drwxrwxrwx 1 root root 4096 Jul 20 2024 'Hacker House Goa Dapp'
drwxrwxrwx 1 root root 4096 Jul 6 2024 'Clone community-classroom-repo'
```

2. List the directory contents size wise sorted.(man ls).

Answer :

- Largest first (descending by size):

Bash : ls -ls

- Smallest first (ascending by size):

Bash : ls -lsr

```

root@PRAKHAR-96378-D1:/mnt/d/Documents/Learning/CDAC# ls
DAC-Feb-2026-Syllabus-and-Marking-Scheme.pdf Hackathon1_c++ Module1_CPP Module2_DBT Module3_Web Module4_OopsJava Module5_DSA Module6_Linux Preparatory_Assignment
root@PRAKHAR-96378-D1:/mnt/d/Documents/Learning/CDAC# ls -ls
total 12100
-rwxrwxrwx 1 root root 12389100 Apr  8 22:56 DAC-Feb-2026-Syllabus-and-Marking-Scheme.pdf
drwxrwxrwx 1 root root 4096 Mar 21 17:22 Hackathon1_c++
drwxrwxrwx 1 root root 4096 Apr  4 00:49 Module1_CPP
drwxrwxrwx 1 root root 4096 Apr  4 00:54 Module2_DBT
drwxrwxrwx 1 root root 4096 Apr 12 16:28 Module3_Web
drwxrwxrwx 1 root root 4096 May 17 20:19 Module4_OopsJava
drwxrwxrwx 1 root root 4096 May 17 23:25 Module5_DSA
drwxrwxrwx 1 root root 4096 May 19 16:26 Module6_Linux
drwxrwxrwx 1 root root 4096 Feb 23 10:42 Preparatory_Assignment
root@PRAKHAR-96378-D1:/mnt/d/Documents/Learning/CDAC# ls -lsr
total 12100
drwxrwxrwx 1 root root 4096 Feb 23 10:42 Preparatory_Assignment
drwxrwxrwx 1 root root 4096 May 19 16:26 Module6_Linux
drwxrwxrwx 1 root root 4096 May 17 23:25 Module5_DSA
drwxrwxrwx 1 root root 4096 May 17 20:19 Module4_OopsJava
drwxrwxrwx 1 root root 4096 Apr 12 16:28 Module3_Web
drwxrwxrwx 1 root root 4096 Apr  4 00:54 Module2_DBT
drwxrwxrwx 1 root root 4096 Apr  4 00:49 Module1_CPP
drwxrwxrwx 1 root root 4096 Mar 21 17:22 Hackathon1_c++
-rwxrwxrwx 1 root root 12389100 Apr  8 22:56 DAC-Feb-2026-Syllabus-and-Marking-Scheme.pdf
root@PRAKHAR-96378-D1:/mnt/d/Documents/Learning/CDAC#

```

3. List directory contents along with their inode no. (man ls).

Answer: Bash : ls -li

```

root@PRAKHAR-96378-D1:/mnt/d/Documents/Learning/CDAC# ls -li
total 12100
2533274790724444 -rwxrwxrwx 1 root root 12389100 Apr  8 22:56 DAC-Feb-2026-Syllabus-and-Marking-Scheme.pdf
78812993479211521 drwxrwxrwx 1 root root 4096 Mar 21 17:22 Hackathon1_c++
844424930457281 drwxrwxrwx 1 root root 4096 Apr  4 00:49 Module1_CPP
1125899907166465 drwxrwxrwx 1 root root 4096 Apr  4 00:54 Module2_DBT
1407374883879413 drwxrwxrwx 1 root root 4096 Apr 12 16:28 Module3_Web
3096224744143324 drwxrwxrwx 1 root root 4096 May 17 20:19 Module4_OopsJava
2533274790905819 drwxrwxrwx 1 root root 4096 May 17 23:25 Module5_DSA
1970324837316630 drwxrwxrwx 1 root root 4096 May 19 16:26 Module6_Linux
562949953746596 drwxrwxrwx 1 root root 4096 Feb 23 10:42 Preparatory_Assignment
root@PRAKHAR-96378-D1:/mnt/d/Documents/Learning/CDAC#

```

4. List the contents of the subdirectory.

Answer : ls subdirectory_name/

Bash : ls Module1_CPP/

```

root@PRAKHAR-96378-D1:/mnt/d/Documents/Learning/CDAC# ls
DAC-Feb-2026-Syllabus-and-Marking-Scheme.pdf Module1_CPP Module3_Web Module5_DSA Preparatory_Assignment
Hackathon1_c++ Module2_DBT Module4_OopsJava Module6_Linux
root@PRAKHAR-96378-D1:/mnt/d/Documents/Learning/CDAC# ls Module1_CPP/
Org_Github Practice_Code Sir_Github
root@PRAKHAR-96378-D1:/mnt/d/Documents/Learning/CDAC#

```

5.1 Create a file, write your surname and name in it and save.

Answer: Bash : touch myfile.txt

Bash : myfile.txt

Bash : cat myfile.txt

Prakhar Vishwakarma

```
prakhar@PRAKHAR-96378-D1:~/temp$ touch myfile.txt
prakhar@PRAKHAR-96378-D1:~/temp$ echo "Prakhar Vishwakarma" > myfile.txt
prakhar@PRAKHAR-96378-D1:~/temp$ cat myfile.txt
Prakhar Vishwakarma
prakhar@PRAKHAR-96378-D1:~/temp$ █
```

5.2 : now open the same file and add your address in it and save.

Bash : `echo "Indore, Madhya Pradesh" >> myfile.txt`

5.3 : reopen the same file and check your name and address in it.

Bash : `cat myfile.txt`

```
prakhar@PRAKHAR-96378-D1:~/temp$ echo "Indore, Madhya Pradesh" >> myfile.txt
prakhar@PRAKHAR-96378-D1:~/temp$ cat myfile.txt
Prakhar Vishwakarma
Indore, Madhya Pradesh
prakhar@PRAKHAR-96378-D1:~/temp$
```

6. how to change the time stamp of file (man touch).

Answer :

```
prakhar@PRAKHAR-96378-D1:~/temp$ cat myfile.txt
Prakhar Vishwakarma
Indore, Madhya Pradesh
prakhar@PRAKHAR-96378-D1:~/temp$ ls
myfile.txt
prakhar@PRAKHAR-96378-D1:~/temp$ ls -l myfile.txt
-rw-rw-r-- 1 prakhar prakhar 43 May 19 18:31 myfile.txt
prakhar@PRAKHAR-96378-D1:~/temp$ touch -t 202605191830 myfile.txt
prakhar@PRAKHAR-96378-D1:~/temp$ ls -l myfile.txt
-rw-rw-r-- 1 prakhar prakhar 43 May 19 18:30 myfile.txt
prakhar@PRAKHAR-96378-D1:~/temp$ █
```

7. Create directory structure : Give following commands.

Answer:

```

prakhar@PRAKHAR-96378-D1:~/temp$ mkdir one
prakhar@PRAKHAR-96378-D1:~/temp$ cd one
prakhar@PRAKHAR-96378-D1:~/temp/one$ touch 1.txt 11.txt 111.txt
prakhar@PRAKHAR-96378-D1:~/temp/one$ mkdir two
prakhar@PRAKHAR-96378-D1:~/temp/one$ cd two
prakhar@PRAKHAR-96378-D1:~/temp/one/two$ touch 2.txt 22.txt 222.txt
prakhar@PRAKHAR-96378-D1:~/temp/one/two$ mkdir three
prakhar@PRAKHAR-96378-D1:~/temp/one/two$ cd three
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three$ touch 3.txt 33.txt 333.txt
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three$ mkdir four; cd four
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four$ touch 4.txt 44.txt 444.t
xt; mkdir five
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four$ cd five
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four/five$ touch 5.txt 55.txt
555.txt
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four/five$ cd ~
prakhar@PRAKHAR-96378-D1:~$ ls
temp
prakhar@PRAKHAR-96378-D1:~$ cd temp
prakhar@PRAKHAR-96378-D1:~/temp$ ls
myfile.txt  one
prakhar@PRAKHAR-96378-D1:~/temp$

```

8. Considering above directory structure , Do following operations :

i.) list the directory contents of directory named "five" from current directory (i.e. home directory).

Answer : `ls ~/temp/one/two/three/four/five`

ii.) write your name in a file named "44.txt" of directory "four" from current directory.

Answer : `echo "Prakhar Gupta" >> ~/temp/one/two/three/four/44.txt`

`cat ~/temp/one/two/three/four/44.txt`

iii.) remove the file named "555.txt" from directory "five" from current directory.

Answer : `rm ~/temp/one/two/three/four/five/555.txt`

`ls ~/temp/one/two/three/four/five`

iv.) change directory to five (i.e. cd one/two/three/four/five/).

Answer : `cd ~/temp/one/two/three/four/five/`

v.) write your course name in file named "3.txt" which resides in directory "three" from current directory (i.e. five).also read the same file.

Answr : echo "PGCP-AC (CDAC)" > ../../3.txt

cat ../../3.txt

vi.) list the contents of directory "two" from current directory (i.e. five).

Answer : ls ~/temp/one/two

```
prakhar@PRAKHAR-96378-D1:~/temp$ mkdir one
prakhar@PRAKHAR-96378-D1:~/temp$ cd one
prakhar@PRAKHAR-96378-D1:~/temp/one$ touch 1.txt 11.txt 111.txt
prakhar@PRAKHAR-96378-D1:~/temp/one$ mkdir two
prakhar@PRAKHAR-96378-D1:~/temp/one$ cd two
prakhar@PRAKHAR-96378-D1:~/temp/one/two$ touch 2.txt 22.txt 222.txt
prakhar@PRAKHAR-96378-D1:~/temp/one/two$ mkdir three
prakhar@PRAKHAR-96378-D1:~/temp/one/two$ cd three
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three$ touch 3.txt 33.txt 333.txt
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three$ mkdir four; cd four
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four$ touch 4.txt 44.txt 444.txt; mkdir five
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four$ cd five
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four/five$ touch 5.txt 55.txt 555.txt
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four/five$ cd ~
prakhar@PRAKHAR-96378-D1:~$ ls
temp
prakhar@PRAKHAR-96378-D1:~$ cd temp
prakhar@PRAKHAR-96378-D1:~/temp$ ls
myfile.txt  one
prakhar@PRAKHAR-96378-D1:~/temp$ ls ~/temp/one/two/three/four/five
5.txt  55.txt  555.txt
prakhar@PRAKHAR-96378-D1:~/temp$ echo "Prakhar Gupta" >> ~/temp/one/two/three/four/44.txt
prakhar@PRAKHAR-96378-D1:~/temp$ cat ~/temp/one/two/three/four/44.txt
Prakhar Gupta
prakhar@PRAKHAR-96378-D1:~/temp$ rm ~/temp/one/two/three/four/five/555.txt
prakhar@PRAKHAR-96378-D1:~/temp$ ls ~/temp/one/two/three/four/five
5.txt  55.txt
prakhar@PRAKHAR-96378-D1:~/temp$ cd ~/temp/one/two/three/four/five/
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four/five$ pwd
/home/prakhar/temp/one/two/three/four/five
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four/five$ ls
5.txt  55.txt
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four/five$ cd ~
prakhar@PRAKHAR-96378-D1:~$ ls
temp
prakhar@PRAKHAR-96378-D1:~$ ls ~/temp/one/two
2.txt  22.txt  222.txt  three
prakhar@PRAKHAR-96378-D1:~$ cd ~/temp/one/two/three/four/five/
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four/five$ echo "PGCP-AC (CDAC)" > ../../3.txt
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four/five$ PGCP-AC (CDAC)
-bash: syntax error near unexpected token `CDAC'
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four/five$ ls
5.txt  55.txt
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four/five$ pwd
/home/prakhar/temp/one/two/three/four/five
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four/five$ echo "PGCP-AC (CDAC)" > ../../3.txt
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four/five$ cat ../../3.txt
```

vii.) remove file named "222.txt" which belongs to directory "two" from current directory (i.e. five).

Answer: rm ../../222.txt

ls ../../

viii.) now change the directory to "one".

Answer: cd ../../../../

ix. remove all files only from directory named "five" from current directory (i.e. one).

Answer: `rm ~/temp/one/two/three/four/five/*`

x. remove directory named "five" from current directory (i.e. one)

Answer : `rmdir ~/temp/one/two/three/four/five`

xi. remove whole directory named "four" from current directory (i.e. one).

Answer: `rm -r ~/temp/one/two/three/four`

xii. change to your home directory.

Answer : `cd ~`

```
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four/five$ rm ../../../../222.txt
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four/five$ ls ../../../../
2.txt 22.txt three
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three/four/five$ cd ../../../../
prakhar@PRAKHAR-96378-D1:~/temp/one$ ls
1.txt 11.txt 111.txt two
prakhar@PRAKHAR-96378-D1:~/temp/one$ rm -r ~/temp/one/two/three/four
prakhar@PRAKHAR-96378-D1:~/temp/one$ ls ~/temp/one/two/three
3.txt 33.txt 333.txt
prakhar@PRAKHAR-96378-D1:~/temp/one$ cd ~
prakhar@PRAKHAR-96378-D1:~$
```

9. change the time stamp of file named "3.txt" which resides in directory named "three".(man touch) .

Answer :

```
prakhar@PRAKHAR-96378-D1:~$ cd ~/temp/one/two/three
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three$ touch 3.txt
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three$ touch -t 202605191830 3.txt
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three$ touch -r reference.txt 3.txt
touch: failed to get attributes of 'reference.txt': No such file or directory
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three$ stat 3.txt
  File: 3.txt
  Size: 15          Blocks: 8          IO Block: 4096   regular file
Device: 830h/2096d Inode: 14715         Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/  prakhar)   Gid: ( 1001/  prakhar)
Access: 2026-05-19 18:30:00.000000000 +0530
Modify: 2026-05-19 18:30:00.000000000 +0530
Change: 2026-05-19 19:03:40.591879387 +0530
 Birth: 2026-05-19 18:35:19.474313605 +0530
prakhar@PRAKHAR-96378-D1:~/temp/one/two/three$
```